

GIA H NGO

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Blog: <https://ngohgia.github.io/>

[Github](#)
[Google Scholar](#)

SUMMARY

I'm passionate about creating practical tools, particularly AI-driven solutions that elevate productivity and improve our everyday lives. With years of experience in machine learning research, I've honed my skills in curating and analyzing large, complex biomedical datasets, with my contributions recognized in leading conferences and journals. I've also had the privilege of leading teams in building products from the ground up—taking them from validation to scaling cloud-based platforms and launching web products powered by large language models and other generative AI techniques.

EDUCATION

Cornell University 2018 - 2022
Ph.D. in Electrical and Computer Engineering
Thesis: Synthesis of Brain Images using Deep Learning
Advisor: Mert R. Sabuncu
MICCAI Young Scientist Award

National University of Singapore 2011 – 2015
B.Eng in Electrical Engineering (First Class Honors)
Minor in Technology Innovations & Entrepreneurship, Tel Aviv University

EMPLOYMENT HISTORY

Head of Engineering July 2025 - Now
Klutch AI, Seattle, USA

- Lead overall engineering strategy and execution, guiding architecture, development, and technical roadmap for Klutch platform.
- Build and manage the engineering team, ensuring high-quality delivery, reliability, and alignment between product, AI, and infrastructure.

Founding Engineer 2024 - July 2025
Klutch AI, Seattle, USA

- Lead the development of an AI-native platform for construction and property management
- Develop and deploy real-time pipelines for natural language understanding and responding to multimodal inputs from different stakeholders

Chief Technology Officer 2022 - 2024
GIVE.asia, Singapore

- Develop the company's tech strategy for various stakeholders, including individual donors, fundraisers, and non-profit organizations (NPOs)
- Work with the engineering team to optimize the software stack and infrastructure
- Spearhead new experimental AI-related products for the philanthropic sector

Applied Scientist 2021
Amazon Alexa, Boston, USA

- Explore possible deep learning methods to improve Alexa's Natural Language Understanding module in handling entities appearing in long dialogues
- Propose a joint optimization strategies for improving the accuracy of named entity recognition and coreference resolution on both synthetic and real dialogue data

Software Engineer 2014 - 2018
GIVE.asia, Singapore

- Build the MVP for a crowd-funding platform with Ruby on Rails + AngularJS + MongoDB
- Migrate the tech stack to Scala + VueJS + PostgreSQL to improve the platform's stability

Research Assistant, Clinical Imaging Research Center 2016 - 2017
National University of Singapore, Singapore

- Develop inference algorithms for unsupervised estimation of brain atlases
- Develop new methods to discover cognitive processes affected by psychological disorders

Software Engineer 2014
Project Ray, Tel Aviv, Israel

- Prototype a mobile application for the visually handicapped to localize their positions
- Build a MVP webpage for crowd-sourcing accessibility information of public places

Research Assistant 2013 - 2015
Institute for Infocomm Research, Singapore

- Develop transliteration algorithms augmented with phonology for low-resource languages such as Vietnamese, Cantonese
- Exploit hierarchical structure of logographic characters for better representational learning

AWARDS (SELECTED)

Best Paper Winner: MICCAI Young Scientist Award	2021
Jacobs Scholar Fellowship for PhD study	2018
Best Paper Finalist: INTERSPEECH Conference	2014
ASEAN Scholarship: Full Undergrad Funding at NUS	2011 - 2015
A*STAR Scholarship: Full Secondary and Junior College Funding	2007 - 2010

SKILLS

- Programming: Python, Typescript, Javascript, Scala, Matlab
- Frameworks: Nuxt, VueJS, Play, PyTorch, TensorFlow, Ruby on Rails, TailwindCSS
- Database: PostgreSQL, MongoDB

PUBLIC SOFTWARE

- BrainSurfCNN: surface-based neural network for predicting individual task contrasts from resting-state functional connectivity [\[code\]](#)
- Text2Brain: neural network for synthesizing brain maps from free-form text query [\[code\]](#)
- Author-Topic model for coordinate-based neuroimaging meta-analysis [\[code\]](#)
- Phonology-augmented statistical framework for machine transliteration [\[code\]](#)
- Minimal WYSIWYG editor that supports simple, unnested JSON outputs [\[code\]](#)
- U-Net based on ResNet34 for localizing lung opacities from chest X-Ray [\[code\]](#)
- Photometric stereo to estimate an object's depth map based on pixel brightness [\[code\]](#)

PUBLICATIONS (SELECTED)

Full publication list available in [Appendix](#) or [Google Scholar](#)

1. GH Ngo et al. (2022) Predicting individual task contrasts from resting-state functional connectivity using a surface-based convolutional network. NeuroImage, 248:118849 [\[paper\]](#)[\[code\]](#)
2. GH Ngo & M Nguyen et al. (2021) Text2Brain: Synthesis of Brain Activation Maps from Free-Form Text Query. In International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 605-614 [\[paper\]](#)[\[code\]](#). **Best Paper Winner: MICCAI Young Scientist Award**
3. M Khosla, GH Ngo, K Jamison, A Kuceyeski, MR Sabuncu (2020) Neural encoding with visual attention. Advances in Neural Information Processing Systems, 33:15942-53 [\[paper\]](#). **Oral Presentation**
4. GH Ngo et al. (2019) Beyond consensus: Embracing heterogeneity in curated neuroimaging meta-analysis [\[paper\]](#)[\[code\]](#)
5. GH Ngo et al. (2019) Phonology-augmented statistical framework for machine transliteration using limited linguistic resources. IEEE/ACM Transactions on Audio, Speech, and Language Processing [\[paper\]](#)[\[code\]](#)

TEACHING & PROFESSIONAL SERVICES

Lecturer for Data-driven Decision Making, Advanced Business Analytics, Management Information System courses at VN-UK Institute for Research and Executive Education, University of Danang 2024 - Now

Lecturer for Introduction to Artificial Intelligence (COS30019) class at Swinburne University, Danang 2024

Teaching Assistant for Computer Vision (ECE5470) graduate class at Cornell 2021

Reviewer for NeuroImage, MICCAI, Medical Image Analysis

TALKS

Beacon 2023 , Unknown Frontiers: Harnessing AI for a Better Future Doing Good in the Age of AI	Nov 2023 Singapore
VNUK , Machine Learning Seminar Synthesis of Brain Images Using Deep Learning	Aug 2022 Da Nang, Viet Nam
MIT , Voxel Talk Accurate Prediction of Individual Task Contrasts from Resting-state Functional Connectivity	Mar 2022 Cambridge, MA
INRIA , Parietal Team Text2Brain: Synthesis of Brain Activation Maps from Free-Form Text Query	Dec 2021 Paris, France
Massachusetts General Hospital , Smoller Lab Synthesis of Brain Maps using Deep Learning	Nov 2021 Boston, MA
Stanford , Computational Neuroimage Science Laboratory Text2Brain: Synthesis of Brain Activation Maps from Free-Form Text Query	Nov 2021 Stanford, CA
Stanford , CS523 Research Seminar in Computer Vision and Healthcare Towards holistic encoding models for predicting fMRI responses to multimodal naturalistic stimuli	Apr 2021 Cambridge, MA
OHBM , Symposium Coordinate-Based Meta-analysis: From Consensus to Discovery Science	Jun 2017 Vancouver, Canada

APPENDIX: PUBLICATION LIST

Source: https://scholar.google.com/citations?user=iKBqU_IAAAAJ&hl=en

JOURNALS

1. GH Ngo, M Nguyen, K Jamison, A Kuceyeski, MR Sabuncu (2022) Predicting individual task contrasts from resting-state functional connectivity using a surface-based convolutional network. *NeuroImage*, 248:118849
2. M Khosla, GH Ngo, K Jamison, A Kuceyeski, MR Sabuncu (2021). Cortical response to naturalistic stimuli is largely predictable with deep neural networks. *Science Advances*, 7:eabe7547
3. M Khosla, K Jamison, GH Ngo, A Kuceyeski, MR Sabuncu (2019). Machine learning in resting-state fMRI analysis. *Magnetic resonance imaging*, 64:101-121
4. GH Ngo, SB Eickhoff, M Nguyen, G Sevinc, PT Fox, RN Spreng & BTT Yeo (2019) Beyond consensus: Embracing heterogeneity in curated neuroimaging meta-analysis. *NeuroImage*, 200:142-158
5. GH Ngo, M Nguyen, NF Chen (2019). Phonology-augmented statistical framework for machine transliteration using limited linguistic resources. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 27: 199-211
6. M Nguyen, GH Ngo, NF Chen (2019) Hierarchical character embeddings: Learning phonological and semantic representations in languages of logographic origin using recursive neural networks. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 28:461-473
7. J Wu, GH Ngo, D Greve, J Li, T He, B Fischl, SB Eickhoff, BTT Yeo (2018) Accurate non-linear mapping between MNI volumetric and FreeSurfer surface coordinate systems. *Human Brain Mapping*, 39:3793-3808

FULL LENGTH REFEREED CONFERENCE PAPERS

1. GH Ngo, M Nguyen, NF Chen, MR Sabuncu (2021) Text2Brain: Synthesis of Brain Activation Maps from Free-Form Text Query. International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 605-614 ([Best Paper Winner: MICCAI Young Scientist Award](#))
2. M Nguyen, GH Ngo, NF Chen (2021) Domain-Shift Conditioning Using Adaptable Filtering Via Hierarchical Embeddings for Robust Chinese Spell Check. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 29:2027-2036
3. GH Ngo, M Khosla, K Jamison, A Kuceyeski, MR Sabuncu (2020) From connectomic to task-evoked fingerprints: Individualized prediction of task contrasts from resting-state functional connectivity. *MICCAI*, 62-71 ([Oral](#))
4. M Khosla, GH Ngo, K Jamison, A Kuceyeski, MR Sabuncu (2020) Neural encoding with visual attention. *Advances in Neural Information Processing Systems*, 33:15942-53 ([Oral](#))
5. M Khosla, GH Ngo, K Jamison, A Kuceyeski, MR Sabuncu (2020) A shared neural encoding model for the prediction of subject-specific fMRI response. *MICCAI*, 539-548
6. M Nguyen, GH Ngo, NF Chen (2018) Multimodal neural pronunciation modeling for spoken languages with logographic origin. *Proceedings of Empirical Methods in Natural Language Processing (EMNLP)*

7. M Nguyen, GH Ngo, NF Chen (2019) Isolating the Effects of Modeling Recursive Structures: A Case Study in Pronunciation Prediction of Chinese Characters. In Proc 2019 Workshop on Widening NLP, 95-97
8. S Singhanian, M Nguyen, GH Ngo, Chen NF (2018) Statistical machine transliteration baselines for news 2018. In Proc Seventh Named Entities Workshop 2018, 74-78
9. GH Ngo, SB Eickhoff, PT Fox, BTT Yeo (2016) Collapsed variational bayesian inference of the author-topic model: application to large-scale coordinate-based meta-analysis. In Proc Int Workshop on Pattern Recognition in Neuroimaging (PRNI) ([Oral](#))
10. M Nguyen, GH Ngo, Chen NF (2016) Regulating orthography-phonology relationship for English to Thai transliteration. In Proc Sixth Named Entity Workshop, 2016:83-87
11. GH Ngo, NF Chen, M Nguyen, B Ma, H Li (2015) Phonology-augmented statistical transliteration for low-resource languages. In Sixteenth Annual Conference of the International Speech Communication Association (INTERSPEECH) ([Best Paper Finalist](#)).
12. GH Ngo, NF Chen, S Sivadas, B Ma, H Li (2014) A Minimal-Resource Transliteration Framework for Vietnamese. INTERSPEECH
13. NF Chen NF, S Sivadas, BP Lim, GH Ngo, H Xu, B Ma, H Li (2014) Strategies for Vietnamese keyword search. InProc IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 4121-4125)